

On the binary format of functional features

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Abstract

In this paper, I argue that the features of functional categories universally conform to a specific equipollent binary format that I will call the Contrary Format, to distinguish it from the more usual Contradictory Format that is usually and implicitly assumed in equipollent binary +/- features or unary features. The term ‘contrary’ adopts Horn’s (1989) distinction between contrary and contradictory negation. To borrow an image, while contradictory values oppose black and white, contrary values oppose white (or black) to every shade of gray. I argue that the Contrary Format functions as a necessary condition for the types of concepts that are expressed as functional categories, and that this perspective on the nature of functional features allows for the discovery of nontrivial relations across grammatical categories. The paper is largely programmatic in nature, and therefore paints in broad strokes rather than providing the fine-grained detail that might be customarily expected.

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1. The Contrary Format

Traditionally, the equipollent opposition between plus and minus values of features is viewed as a polar contrast between the presence and absence of a given feature value when both are grammatically active. A typical example from phonology would be [$\pm$ Voice], with Voice being either present or absent (Cowper & Hall 2014). Privative monovalent features are distinguished by the fact that only one pole of the opposition is grammatically active. Cowper & Hall (2014) propose to indicate the absence of such a feature as [non-FEATURE]. In both cases, the opposition is a polar one: ‘non’ and ‘minus’ represent the polar complement of the positive feature value.

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Thanks to Camil Staps for many discussions and excellent questions, and for drawing my attention to Dixon (2021), and to Pierre Pica for the inspiration provided by our ongoing and now decade-long conversation about the Mundurucu number system and classifiers. *Ars longa, vita brevis.*

In Horn's (1989) terms, 'plus/ minus' feature values are an instance of contradictory negation. Horn (1989) notes that 'contradictory' values have a meaning that involves the complement set of the positive value: the 'contradictory' negation *non* in *non-American* refers to any nationality or property that is not American, while the 'contrary' negation expressed by *un-* in *un-American* refers to something that has a very low level of characteristics that are typically associated with America. Similarly, *non-christian* refers to the complement set of 'christian', while *unchristian* can easily be predicated of someone who identifies as a Christian: they just display the lowest possible degree of Christian values.

I would like to propose that – at least for functional categories in morphosyntax² – the format of the equipollent binary opposition is 'contrary' in nature rather than 'contradictory'. The positive values in this 'contrary' binary format are *distinct*, *discontinuous*, or *well-delineated*. By contrast, their negative values are of the contrary *un-* type rather than of the contradictory *non-* type. Rather than the mere absence of a feature value, the negative values of functional features in morphosyntax indicate an *indistinct*, *imprecise*, *continuous*, or *not well delineated* value for that feature. To make this more clear, it can be useful to rephrase color oppositions in terms of contrary and contradictory values: while contradictory values oppose black and white, contrary values oppose white (or black) to every shade of gray.

In other words, the values in opposition in the Contrary Format each have a particular signature that is not reducible to the presence or absence of a value. The negative value of functional features has a semantically much richer structure than the mere absence of that feature can account for. I will argue that this richer structure accounts for diachronic and synchronic relations between the negative contrary values of at first sight very different functional categories that could not be explained by the mere absence of their positive values.

Finally, I will argue that functional categories must conform to the properties of the Contrary Format to be expressible at all. It is my contention that the feature value oppositions inherent in functional categories must either express a Contrary Format, or be compositionally built with Contrary Format feature values. Concepts that do not fit the Contrary Format cannot be expressed as functional features.

2. Discussion

A minute and encyclopedic demonstration of the Contrary Format for each functional category could quickly become tedious. I will therefore succinctly discuss how the Contrary Format manifests in the nominal, verbal, and even in the prepositional domain. I will consider some more difficult cases in section 4. I acknowledge that many of the roughshod and perhaps somewhat elliptical assertions I will be making are in need of considerable refinement and elaboration. I therefore apologize in advance to the reader for any irritation that may arise in the reading of this paper, which is intended and should be read as a programmatic sketch rather than as a full-fledged proposal.

² Of course this immediately raises the question why the binary opposition in phonology would not be 'contrary' instead of 'contradictory'. I will not address this question here. Perhaps phonology is really different (Bromberger & Halle 1989).

2.1. The nominal and adjectival domain.³

The Contrary Format can perhaps most clearly be observed in the **definite/ indefinite** distinction in the determiner domain: definite nouns are identifiable and identified in context, while indefinites are either not yet identified or not quite identifiable. Note that if indefinites were to be characterized as ‘non-identifiable’ in the ‘contradictory’ sense, they should in principle be ‘closed off’, or simply inaccessible in the subsequent discourse. That is of course not what indefinites do: the reference of a typical indefinite can be picked up and further identified in the following discourse (2a). Contrast the indefinite article *a* with *some*, which does not allow its complement to be referred to in subsequent discourse in the same way, as in (2b).

- (2) a. *I read a book. The book was great.*
 b. *? I read some book. The book was great.*

The meaning of the indefinite article *a(n)* therefore does not refer to the negated polar opposite of the meaning of the definite article. Instead, the indefinite article expresses a different, less distinct notion of identifiability: not quite identifiable or unidentified, rather than impossible to identify, or ‘non-identifiable/ identified’. Indefinites refer to rather well circumscribed sets containing individuals that are not individually further identified, and therefore can sometimes be randomly selected from their class. A singular indefinite can refer to any member of its class. That does not mean it is characterized by a complete absence of definiteness. Rather, it is not yet fully identified. Similarly, quantifiers like *many*, *some*, and *few* can be viewed as variations on indefinite determiners. They represent expressions of magnitudes that fall on the continuous side of the Contrary Format.

Demonstratives are characterized by the proximal/ distal opposition. As noted by Woodworth (1991) and Diessel (1999), the proximal/ distal opposition is often associated with *visible/ invisible* (or *known/ unknown*): proximal entities can be made out clearly, while distal ones are less visible and distinct, and are situated on a spectrum of visibility. The notion Proximal defines a specific space close to the Speaker. Medial is an intermediate deictic category, that can be said to be indirect and removed from the Speaker, but direct and close for the Hearer. By contrast, Distal does not indicate a specific space, but can be used to indicate a rather large spatial or contextual range for both Speaker and Hearer. To conclude, I take Proximal to represent the ‘distinct’ value of the Contrary Format, while Distal represents its ‘indistinct’ value.⁴

In the nominal domain, the distinction in **Number** between singular and plural can be viewed as a contrary opposition, with singular as the discrete and discontinuous value of the feature, and plural as its non-discrete, continuous counterpart: a plural is typically a continuous quantity greater than one, a magnitude. The opposition between singular and plural could also be viewed as an opposition between [individual] and [nonindividual]. In addition, I view duals (and perhaps trials) as Number features that are separate and distinct from the singular/ plural opposition. Duals often have two values – discrete and non-discrete – that conform to the Contrary Format, as evidenced by the fact that duals can exhibit both discrete (‘exactly

³ Gender, a more recalcitrant case, will be discussed in Section 4.

⁴ Diessel (1999:44) also mentions *upriver/ downriver/ across river* and *uphill/ downhill*, which seem to involve a contradictory opposition rather than a contrary one. However, such elements are composite and take a landmark (*hill, river*) as a definite reference point for calculating an indefinite direction or location. See also other cases of composite grammatical elements below, including *Aktionsarten* and prepositions.

2') and approximate, continuous, paucal ('more or less two, a few') interpretations, and that paucals historically often evolve from duals.⁵

(1)

Number	discrete	non-discrete
Individual	singular	plural
Dual	exactly 2	about 2/ paucal

Note also that the lexical counterparts of duals, like *pair* and *couple* in English, show great variety across languages in affording discrete or approximate interpretations. For example, while *pair* is strictly discrete in English (*a pair of shoes/ ??nails*), its counterpart *paar* 'pair' in Dutch can be either discrete (*een paar schoenen* 'a pair of shoes') or approximate (*een paar spijkers* 'a few nails').

The category **Person** represents more of a challenge. 1/2P represent the conversation Participants: they are proximal and distinct. By contrast, 3P is usually viewed as the 'nonparticipant' in a 'contradictory' manner (Forchheimer 1953:5-6). Harley & Ritter (2002) therefore represent Speaker and Addressee (1/2P) as dependents of a PARTICIPANT node, while 3P is the 'non-participant' and depends on a separate INDIVIDUATION node. In other words, as the 'non-participant' or mere 'absence' of participation in a conversation, 3P represents a 'contradictory' value, not a 'contrary' one. I would like to challenge this perspective. Harley & Ritter's (2002) person geometry predicts that 1/2P forms should be similar to each other but different from 3P forms. However, that is rarely the case: in most languages, 1/2/3P are integrated into the same morphological paradigm, and 2/3P can even share the same exponents. Instead, I would like to propose – in the spirit of the Demonstrative proximal/ medial/ distal relation – that 3P is the 'distal' or 'remote' Person: an entity that can be referred to by Speaker and Addressee, but is a Participant nonetheless. This view is corroborated by the fact that some Semitic languages use 3P pronouns as distal demonstratives. The fact that 3P is 'remote' can also be related to the fact that most languages express the impersonal form as 3P, as pointed out by Benveniste (1946). Impersonal *it* in *It's important to exercise* or *It is raining* is hard to understand as an expression of INDIVIDUATION, but makes perfect sense as a placeholder referring to a 'yet-to-be defined' situation (*to exercise*) or to a meteorological event like *rain*.⁶ This 'yet-to-be-defined' aspect of impersonal pronouns represents a 'contrary' value. I would therefore argue that 3P can be taken to express the 'undefined', indistinct Person value of the Contrary Format.

Case displays a distinction between *structural* and *inherent* case (Chomsky 1986). That opposition conforms to the Contrary Format. Structural case is assigned in a specific,

⁵ Note that this means that plural is not in a paradigmatic opposition with dual or trial. Under this perspective, plural is simply 'non-individual': an unspecified quantity greater than one. The approximate interpretation of duals and trials, by contrast, has by necessity a paucal interpretation. This explains Corbett's (2000:23) observation that "Systems with just a paucal in addition to singular and plural are rare. It is much more common to find it with a dual too". Admittedly, this perspective on Number features is very different from that adopted in e.g. Harley & Ritter (2002) and subsequent work. But their [+Minimal, +Group] definition of dual fails to do justice to the fact that duals are often sensitive to the (non)complementarity of the elements in their set, either requiring two complementary items (e.g. body parts), as in Akkadian, or dispreferring them, as in Slovene. Also note that the discrete/ non discrete sg/ pl opposition in Table 1 is a morphological one, while the discrete/ non discrete interpretations of dual/ trial are semantic in nature. Clearly, this deserves a more extensive inquiry.

⁶ See also Bolinger's (1973) 'ambient' *it*.

well-defined structural configuration: e.g. nominative case is assigned to the argument that agrees with the verb in person and number. By contrast, *inherent* case is dependent on an unspecified, fuzzy, and often idiosyncratic range of thematic roles: dative case can be used for Goal, Experiencer, Location, or Beneficiary; genitive for Possessor and Part-Whole relations etc.

Finally, in the adjectival domain, the functional categories of **Comparison**, i.e. comparative and superlative morphology, can be viewed as expressions of, respectively, the continuous and distinct values of the Contrary Format.

2.2. The verbal and clausal domain

Moving nimbly to the verbal domain, although **Aktionsart** is not technically a functional category, the building blocks of States, Processes, Accomplishments and Achievements reflect the Contrary Format. States represent an indistinct, continuous value, while Achievements represent a discrete, distinct transition from one state to another. Processes and Accomplishments are compositional. Roughly speaking, Processes are continuous events that consist of multiple distinct and discontinuous subevents: *dance, rain, walk*. The Contrary Format captures both States and Processes as continuous magnitudes even though their internal structure is different. Accomplishments are composed of continuous Processes that culminate in a distinct transition.

The same is true for **Aspect**: aspectual categories can be divided into discrete properties (e.g. inchoative, semelfactive, resultative, completive), and continuous ones (e.g. imperfective, progressive, incomplete). Perfect aspect may be viewed as complex: it refers to a (continuous) set of stages of an anterior event, and sometimes implies the (distinct) completion of that prior event. As Comrie (1976:52) notes, perfective aspect looks at a situation ‘as a single whole’ (a distinct value) whereas the imperfective looks at its ‘internal structure’ (a continuous value). Repetitive aspect can be viewed as a combination of continuous and discrete properties: a continuous set of identical discrete events.

Voice is the grammatical category that encodes the relation between an event and its participants. I would like to argue that Voice represents the way in which the Agent is directly or indirectly involved in the event. In the Active voice, the Agent is the subject and presented as being *directly* involved in the event. In the Passive voice, the Agent is presented as less directly involved in the event. In many languages, this more indirect involvement of the Agent is encoded by locative prepositions: *by* in English, *par* ‘via’ and *de* ‘from’ in French, *von* ‘from’ in German (See Staps 2024). Relational Grammar captured it somewhat imperfectly with the notion that the Agent is ‘demoted’ in the passive. In the Middle voice, the Agent is not involved in the event, and can be said to be entirely indirect or even absent. An analysis of Voice in terms of direct and indirect involvement allows for an interesting comparison with the notions of Proximal, Medial, and Distal in the demonstrative domain: the direct involvement of the Agent in Active voice relates to the directness of the Proximal locative, while the indirect locative prepositions used in the Passive voice indicate a relation with indirect Medial locations. The absence of the Agent in the Middle voice in turn relates to the relative invisibility of Distal locations. I would like to suggest that these associations are not merely metaphorical, but reflect robust relations across grammatical categories, mediated by the Contrary Format.

Similarly, the opposition in **Tense** between present and past (assuming future to be part of modality) can be likened to the proximal/ distal opposition: in a Reichenbachian account, present tense indicates that Situation Time and Reference Time coincide with Speech Time, a very distinct and deictic moment.⁷ By contrast, past tense can refer to any moment on a temporal interval before the present. In many theories of tense, past tense is assumed to involve existential quantification over times. So past tense functions as the ‘unidentified’ value of Tense, in the same way that a singular indefinite can refer to an unidentified individual of a class. Past tense can be used as an indirect evidential in a language like Somali (Lecarme 1996, 1999, 2008). This is in line with Lau & Rooryck’s (2017) work on the relation between perfect aspect and indirect evidentiality (see below). The past and the perfect share with indirect evidentiality a contrary value that can be either temporal/ aspectual, or be interpreted evidentially in terms of continuous information updates.

Verbal **Mood** at least includes the indicative/ subjunctive opposition. The indicative is the mood of well-delineated reality and factuality, while the subjunctive refers to not-well-delineated events that are not actually the case or possible, desirable, and counterfactual. As noted by Portner (2018) citing Quer (1998), in many Romance languages, relative clauses containing the indicative require a specific, extensional interpretation; while relative clauses with the subjunctive are unspecific, in contrary terms, and yield an intensional reading for the noun they modify. The opposition is very similar to that between definite and indefinite in the sense defined above. Once again, the subjunctive has a much richer internal structure than a mere absence of reality or factuality: the subjunctive is used for events whose reality cannot be quite made out or defined, not just for events that lack any reality whatsoever. It has often been observed that infinitives are very close to subjunctives in many languages, and that subjunctives take over the role of infinitives in languages that lack them. Infinitival morphology indicates that an event is not linked to a specific point on the time axis. They can therefore be viewed as indefinite in the relevant sense. Proper credit for this insight goes to Tsoulas (1995, 1996:298), who first argued that the finite/non-finite distinction in clausal complementation can be better described in terms of ‘definite’ and ‘indefinite’ propositions, thus underscoring the relation between the verbal and nominal domains.

When it comes to sentence mood or **Force**, declarative mood clearly is on the definite/ distinct side of the Contrary Format, while interrogative and imperative seem to be on the ‘indefinite’ side. Imperatives are close to the subjunctive in that they refer to a set of desirable situations rather than to an actual situation, and are therefore [irrealis] in the sense of Han (1998). Since Hamblin (1958, 1971, 1973), the semantics of interrogatives has been taken to be the set of possible answers. Since this set is not exhaustively defined, I take it to be compatible with the ‘indefinite’ value for the feature Force.

The category of **Modality** distinguishes between deontic and epistemic modality. In Kratzer’s (1977, 1981, 1991) work, deontic and epistemic modals are uniformly analyzed as quantifiers over possible worlds. The difference between deontic and epistemic modality lies in their modal base. The epistemic modal base draws on the speaker’s personal or general knowledge, while a deontic modal base relates to obligations, rules, and capacities. The epistemic sentence *It must be raining in Paris* is interpreted evidentially as a function of the speaker’s indirect evidence for that statement: the speaker concludes that it is raining in Paris from personal knowledge or via hearsay, the weather report, etc. The exact source of knowledge on

⁷ Admittedly, present tense can be used to refer to future events and habitual/ generic ones. But as noted by Smith (2008), the interpretation as present is the default.

Moreover, direct evidentials indicate that the Speaker presents the proposition as if they were present at the event, while indirect evidentials suggest that the speaker was not present at the event and came to know it by indirect means. Rooryck (2019) argues that the distinction between direct and indirect evidentiality is similar to the proximal/ distal distinction in demonstratives. A direct evidential marks that the event was located close to the Speaker, while an indirect evidential indicates that the Speaker was far away from the event. I argue that this parallel between locative and evidential meanings is facilitated by the Contrary Format, since it aligns direct evidentiality with proximal location on the one hand, and indirect evidentiality with distal location on the other.

More importantly, the Contrary Format for evidential markers allows for an explanation of the relationship between present perfect morphology and indirect evidentiality. In many languages (e.g. Turkish, Bulgarian, and Norwegian) present perfect morphology is ambiguous between an aspectual interpretation and an interpretation in terms of indirect evidentiality (inference or hearsay) (Izvorski 1997, Lau & Rooryck 2017). Lau & Rooryck (2017) account for this via semantic underspecification. At their invariant core, both perfect aspect and indirect evidentiality involve a template that refers to a set of stages. Perfect aspect involves event stages, while indirect evidentiality interprets these stages in terms of information stages: inference and hearsay. Indirect evidential markers indicate that the final information stage presented in *p* was arrived at by a Speaker-external information update process of various sources telling them that *p* (hearsay), or by a Speaker-internal information update process of inference on the basis of general knowledge or other clues. In terms of the Contrary Format, the contrary negative value represented by perfect aspect and indirect evidentiality functions as a semantically rich enough scaffold that allows to link the aspectual and evidential interpretations.

Note that it would be impossible to derive this relation between perfect aspect and indirect evidentiality from the mere negative value or absence of the positive counterparts of perfect aspect or indirect evidentiality. This is simply because it is not possible to make anything follow from the absence of something, to slightly reinterpret Parmenides' dictum *οὐδὲν ἐξ οὐδενός* 'Nothing comes from nothing' (Latin: *ex nihilo nihil fit*).

2.3. Prepositions

Perhaps most surprisingly, prepositions also exhibit properties of the Contrary Format. Dixon (2021:11) notes that English prepositions can be divided into what he calls major and minor members. Major members include *to*, *in*, *on*, *up*, *over*, *above* and *after*. These are paired with minor members: *from*, *out*, *off*, *down*, *under*, *below*, and *before*. In these lists, the major member is more frequent than the minor member, has a wider range of meanings than the minor member, and has a positive specification that is often roughly the opposite of the minor member.

At least for some of the preposition pairs contained in these sets, the nature of the opposition between them can be formulated in terms of the Contrary Format. *In* and *on* are spatially bounded and distinct, while *out* and *off* lack distinct spatial boundaries. Note that *out* and *off* do not denote the mere absence of spatial boundaries with respect to containment or location. The category denoting the absence of spatial boundaries is much larger, and includes additional indistinct locations including *about* and *around*, that are quite different from *out* and *off*. *About* and *around* can be viewed as instantiations of a contrary value rather than a contradictory one, indicating a space lacking specific boundaries surrounding a specific

location. The ‘distinct’ value at the center of this specific location is arguably represented by the preposition *at*.

The oppositions between *to* and *from*, *up* and *down*, *over* and *under*, *above* and *below*, and *before* and *after* cannot be reduced to the Contrary Format. However, they are all composed of a distinct value and a less distinct one. All of these prepositions define an indistinct direction, location, or time as a function of a distinct location or time. I would therefore like to propose that they are semantically composed of both distinct and indistinct values of the Contrary Format, and are therefore very similar to the composition of distinct and indistinct values in Aktionsart and Aspect as discussed above.

3. A few recalcitrant cases

After a whirlwind tour of functional (and semi-functional) categories, I will consider the hardest nut to crack for the Contrary Format: **Gender**. It is clear that the traditional values for the gender of lexical nouns – masculine, feminine, and neuter – cannot be interpreted in terms of the Contrary Format. However, Gender is one of the least understood functional categories. Animate nouns are called masculine or feminine as a function of the biological sex of the animate entities they refer to. As Dahl (2000: 102) puts it: “*The pervasiveness of sex as gender criterion is striking. There are many possible ways of classifying animates, in particular human beings, that might be used as a basis for gender, such as social status, ethnic origin, profession, age, hair color, etc., but none of them except perhaps age seems to play any important role in gender assignment.*” However, inanimate nouns also display masculine and feminine gender in most languages, often without a correspondence to biological sex. The gender of such nouns is arbitrary: Italian *mare* ‘sea’ is masculine and French *mer* ‘sea’ is feminine, and that just has to be stipulated in the lexicon of each language for each inanimate noun. I will not be able to address here why animacy, gender and biological sex are correlated in Gender.

To gain an insight in Gender in terms of the Contrary Format, I will turn to cases where Gender is strictly functional, as in derivational affixes and pronouns. Rooryck & Staps (in prep.) develop an analysis of functional gender in terms of *differentiation* and *interlinking*. This allows them to analyze feminine singulative morphemes in Hebrew, Breton and Welsh as the result of an operation of *differentiation*. Applying feminine singulative *-a* to the unmarked masculine collective */bordgen/* ‘orange.MASC.COLL (orange)’ vs. in Tunisian Arabic yields singulative */bordgen-a/* ‘orange-FEM.SING (one orange)’ (Dali & Mathieu 2021: 18). For Rooryck & Staps (in prep.), this is because the singulative *differentiates* among the individuals in the collective, thus ‘zooming in’ on a particular individual that is different from other individuals in the collective. This means that the feature value ‘feminine’ can be reinterpreted in terms of the distinct feature of differentiation. The same is true for feminine pronouns in Dutch. As first observed by Simons (1919), the Dutch feminine pronoun *zij* ‘she’ can refer to ‘internally complex’ words like *regering* ‘government’. Rooryck & Staps (in prep.) reinterpret this observation in terms of differentiation and interlinking: ‘government’ is conceived of as an *interlinked* set of *differentiated* branches that complement each other. Simons (1919) also observes that the neuter pronoun *het* ‘it’ can refer to entities with a very low level of *eenheidsgraad* ‘degree of unity’, as he calls it: materials like old iron, and events like fire. For Staps & Rooryck (in prep.), this is because neuter is [+differentiated] but [–interlinked]: *old iron* can refer to a set of differentiated but otherwise unrelated objects with the same material property. Similarly, fire has a number of differentiated but not necessarily interlinked properties: heat, smoke, and flames. Finally,

masculine pronouns like *hij* ‘he’ can be used to refer to entities that are analyzed as internally *undifferentiated* and *uninterlinked* units. A reinterpretation of masculine, feminine, and neuter in terms of *differentiation* and *interlinking* is in line with the Contrary Format, since both of these operations (or feature values) have the property of distinctness, while the lack of differentiation and interlinking correlates with the contrary values of *indistinct*, *imprecise*, and *not well delineated*.

Classifiers represent another interesting case, from a very specific theoretical perspective. In Rooryck (2024), I propose that classifiers encode the result of a universal, internalist, and computational cognitive mechanism that describes the dynamic patterns of spatial distribution of the material denoted by the predicate in their complement. Instead of the traditional distinction between sortal and mensural classifiers, I propose a more fundamental distinction in the classifier domain between Samples and Pointers, each with distinct properties in terms of countability and spatial distribution. ‘Countable’ Samples (e.g. *splash*, *drop*, *slice*, *spoonful*) *directly* describe how specific materials distribute in space, while ‘uncountable’ Pointers (e.g. *trace*, *whiff*, *tinge*, *note*) *indirectly* encode spatial distribution. Samples operate through a mechanism of identity: a Sample is materially representative, identical to, and typical of the material sampled. Pointers operate via a mechanism of functional similarity: a Pointer does not need to be identical to the material they represent. From this description, it is clear that Samples represent the ‘defined’, ‘direct’, or ‘distinct’ value of the Contrary Format, while Pointers represent its ‘undefined’, ‘indirect’, and ‘less distinct’ value.

Finally, what about discourse information features like the **Topic vs Focus** partition? Focus encodes discourse-new or contrastive information, while Topic encodes discourse-old information (e.g. Gundel & Fretheim 2006). The elements that are focalized or topicalized can themselves be either specific or nonspecific in reference. But Topic and Focus are not about reference, they concern information structure, i.e. the way in which information is presented in a sentence as old or new. I would like to propose that ‘discourse-old’ and ‘discourse-new’ can be compared to distal and proximal in the locative domain. As argued by Staps & Rooryck (2022), discourse-old information is distal because it is in the Common Ground, known and shared by the Addressee. In this sense, discourse-new information could be viewed as the distinct value of the Contrary Format, while discourse-old information can be associated with its less distinct, contrary value. Admittedly, this somewhat speculative perspective may require a more careful development than I am able to present here.

4. A necessary format

Rizzi & Cinque (2016:155) ask the question why the inventory of functional elements is restricted in precisely the way that it is:

"Why do only certain notions find a grammatical encoding within the verbal or nominal domains in the languages of the world? Cinque & Rizzi (2010a, p. 65), Cinque (2013c), and the authors of the references cited there observe that only a few of the concepts and distinctions that populate our cognition are grammatically encoded through bona fide functional exponents like affixes and other closed-class categories (e.g., auxiliaries, particles, clitics). One could expect many more cognitive notions to be grammatically encoded on some category than what we actually find—for example, the notion of ‘anger’ or ‘worry’ encoded as an affix on common nouns or verbs, the notion ‘sexual attractiveness’ encoded in proper human nouns, or the notion of ‘advantage’ encoded in nouns or spatial prepositions."

Rizzi & Cinque's (2016) question can be reformulated as asking for the necessary and sufficient conditions that must be fulfilled for a given cognitive notion to be able to grammatically operate as a functional category. I would like to propose that emotions, colours, sexual attractiveness, or advantage cannot function as functional categories because they do not fit the Contrary Format. An emotion such as anger, or a color such as yellow cannot – as far as I can see – be decomposed into an opposition between a distinct and an indistinct, contrary value, along the lines of *yellow* and **unyellow*, or *angry* and **unangry*. I argue that functional categories are limited to those notions that can be defined or decomposed in terms of the Contrary Format. In other words, I claim that the Contrary Format is a necessary condition for functional categories: an obligatory template that concepts must fit into to be able to be expressed as functional categories in the grammar. As such, it makes very specific predictions about what notions can and cannot be functional categories in the grammar.

The discussion presented here raises many additional questions, and needs to be refined and elaborated in much more detail. One question is why the contrary value refers both to *indistinct* and *arbitrary* values. For instance, an indefinite plural can be called *indistinct* in reference because the boundaries of the set are not precisely defined. By contrast, a singular indefinite can be called *arbitrary* because it does not need to refer to a distinct individual: any member of the set will do. A more general and more puzzling question is why precisely the Contrary Format, rather than any other format, would serve as the common scaffold for functional categories. I will leave such intriguing issues for further research.

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